

ARCHITECTURE REPORT

1. REQUIREMENTS

```
{
  "system_name": "OnlineFoodOrderingSystem",
  "architecture_style": "Microservices Architecture",
  "functional_requirements": [
    "User registration and login",
    "Browse restaurants and menus",
    "Place food orders",
    "Track order status",
    "Process online payments"
  ],
  "non_functional_requirements": {
    "scalability": "High",
    "availability": "High",
    "latency": "Medium",
    "reliability": "High"
  }
}
```

2. ARCHITECTURE (ACME ADL)

```
System OnlineFoodOrderingSystem {
  Property view : String = "runtime";
  Component UserService = new Component {
    Port in;
    Port out;
    Property responsibility : String = "Handle user registration and login functionality";
    Property state : String = "stateless";
    Property scaling : String = "horizontal";
  }
  Component RestaurantService = new Component {
    Port in;
    Port out;
    Property responsibility : String = "Manage restaurant information and menus";
    Property state : String = "stateless";
    Property scaling : String = "horizontal";
  }
  Component OrderService = new Component {
    Port in;
    Port out;
    Property responsibility : String = "Manage food orders and order status";
    Property state : String = "stateless";
    Property scaling : String = "horizontal";
  }
  Component OrderStatusService = new Component {
```

```

Port in;
Port out;
Property responsibility : String = "Update order status and notify users";
Property state : String = "stateless";
Property scaling : String = "horizontal";
}
Component PaymentService = new Component {
    Port in;
    Port out;
    Property responsibility : String = "Process online payments";
    Property state : String = "stateless";
    Property scaling : String = "horizontal";
}
Attachment UserService.out to OrderService.in {
    Property delivery : String = "at-least-once";
    Property ordering : Boolean = false;
    Property backpressure : String = "bounded";
}
Attachment UserService.out to PaymentService.in {
    Property delivery : String = "at-least-once";
    Property ordering : Boolean = false;
    Property backpressure : String = "bounded";
}
Attachment RestaurantService.out to OrderService.in {
    Property delivery : String = "at-least-once";
    Property ordering : Boolean = true;
    Property backpressure : String = "bounded";
}
Attachment OrderService.out to OrderStatusService.in {
    Property delivery : String = "at-least-once";
    Property ordering : Boolean = false;
    Property backpressure : String = "bounded";
}
Attachment OrderStatusService.out to UserService.in {
    Property delivery : String = "at-least-once";
    Property ordering : Boolean = false;
    Property backpressure : String = "bounded";
}
Attachment OrderService.out to PaymentService.in {
    Property delivery : String = "at-least-once";
    Property ordering : Boolean = false;
    Property backpressure : String = "bounded";
}
Attachment PaymentService.out to OrderStatusService.in {
    Property delivery : String = "at-least-once";
    Property ordering : Boolean = false;
    Property backpressure : String = "bounded";
}
Property resilience_pattern : String = "Retry + CircuitBreaker";
Property failure_isolation : Boolean = true;

```

```

Property authentication : String = "service-to-service";
Property authorization : String = "RBAC";
Property encryption : String = "in-transit";
Property constraint_scalability : String = "High";
Property constraint_availability : String = "High";
Property constraint_latency : String = "Medium";
Property constraint_reliability : String = "High";
}

```

3. C4 CONTAINER DIAGRAM

